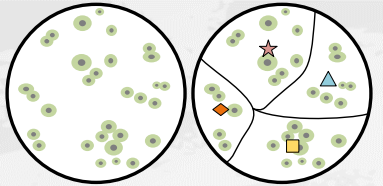

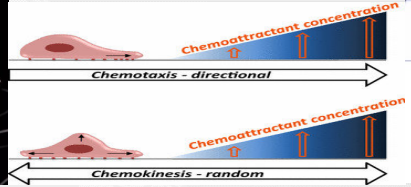

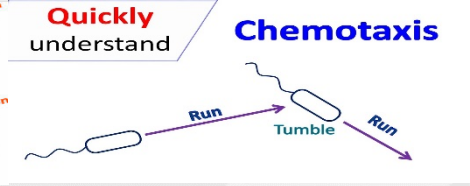

A Population based Bacterial Colony Optimization with Dynamic Multi-Leader Co-evolution for Multi-Objective Optimization Problems

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Management Science and Engineering (2020)

Email: yixinwong1998@gmail.com

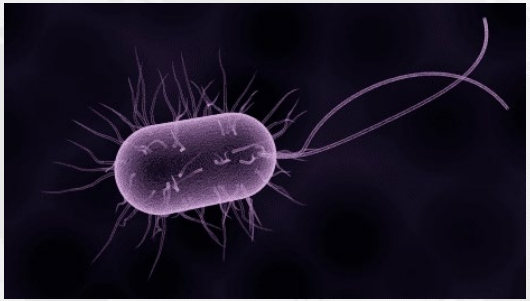
Bacterial Foraging Behaviors

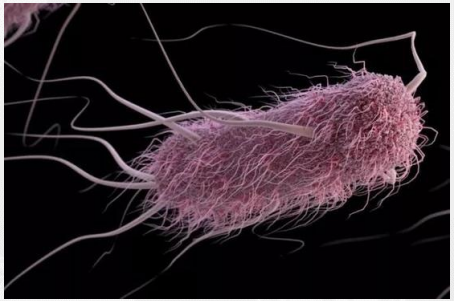
Communication

Elimination

Chemotaxis

Reproduction





Motility typical of flagellated rod-shaped bacteria

Bacterial Foraging Optimization

Chemotaxis: (Tumble; Swimming)

$$\theta^i(j+1,k,l) = \theta^i(j,k,l) + C(i)\frac{\Delta(i)}{\sqrt{\Delta^T(i)\Delta(i)}}$$

Communication:

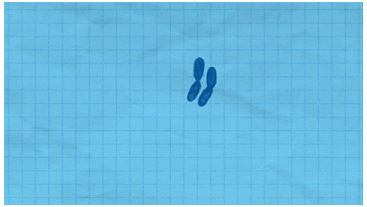
$$J_{cc}(\theta, P(j,k,l)) = \sum_{i=1}^{n} J_{cc}^{i}(\theta, \theta^i(j,k,l))$$

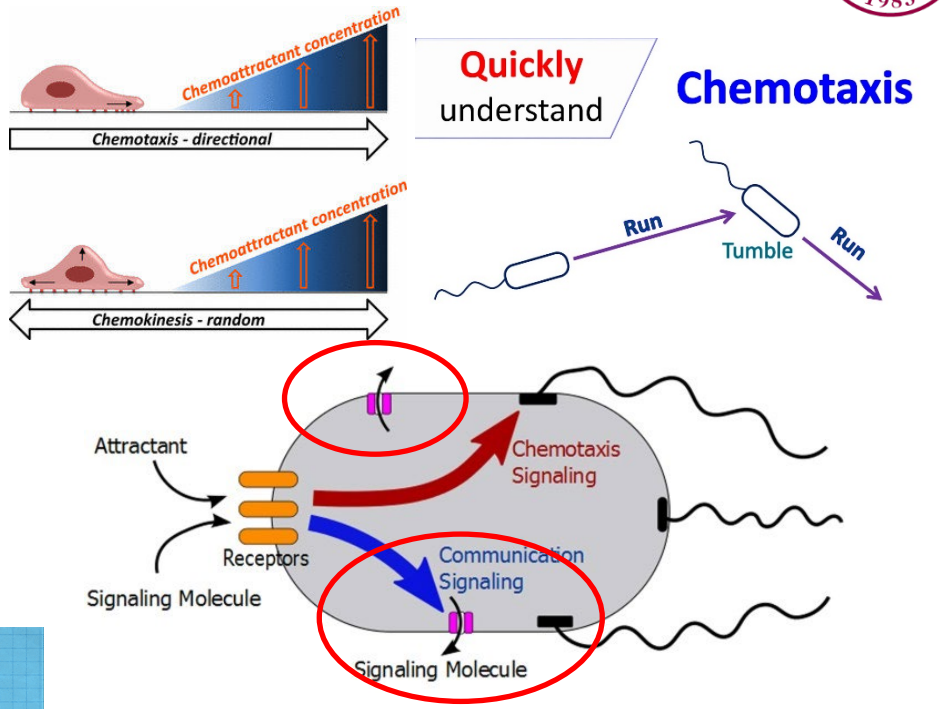
$$+\sum_{i=1}^{n}[h_{repelent}\exp(-w_{repelent}\sum_{m=1}^{p}(\theta_m - \theta_m^i)^2)]$$

$$+\sum_{i=1}^{n}[-d_{attract}\exp(-w_{attract}\sum_{m=1}^{p}(\theta_m - \theta_m^i)^2)]$$

Reproduction:

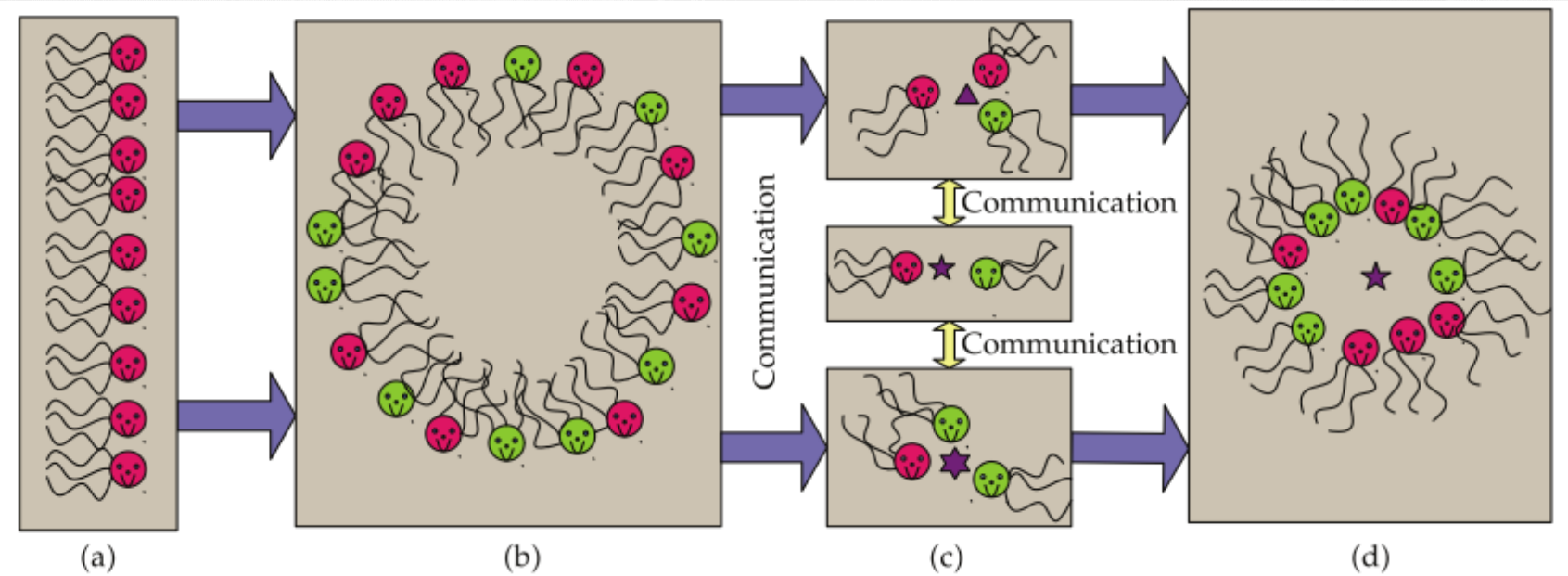
$$J_{health}^{i} = \sum_{j=1}^{N_c+1} J(i,j,k,l)$$





Passino, K. M., 2002, Biomimicry of Bacterial Foraging for Distributed Optimization and Control, IEEE Control Systems Magazine, 22, 52-67.

Bacterial Colony Behavior



Bacterial individual behavior and social behavior

Bacterial Colony Optimization

Information transfer mechanism:

Tumble:

$$\begin{aligned}\text{Poistion}_i(T) = \text{Poistion}_i(T-1) + C(i) \\ * \Big[f_i * (G_{\text{best}} - \text{Position}_i(T-1)) + (1 - f_i) \\ *(P_{\text{best}_i} - \text{Position}_i(T-1)) + \text{turbulent}_i \Big]\end{aligned}$$

Swimming:

$$\begin{aligned}\text{Poistion}_i(T) = \text{Poistion}_i(T-1) + C(i) \\ * \Big[f_i * (G_{\text{best}} - \text{Position}_i(T-1)) + (1 - f_i) \\ *(P_{\text{best}_i} - \text{Position}_i(T-1)) \Big],\end{aligned}$$

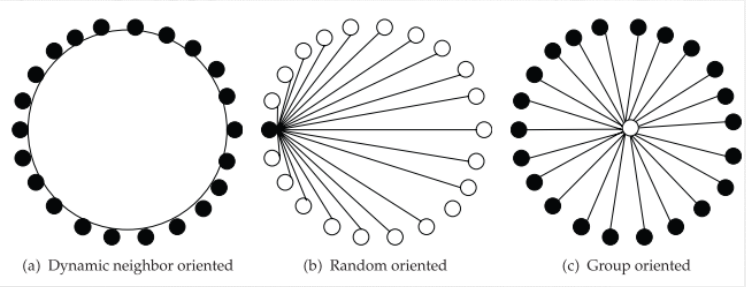
Adaptive Step size:

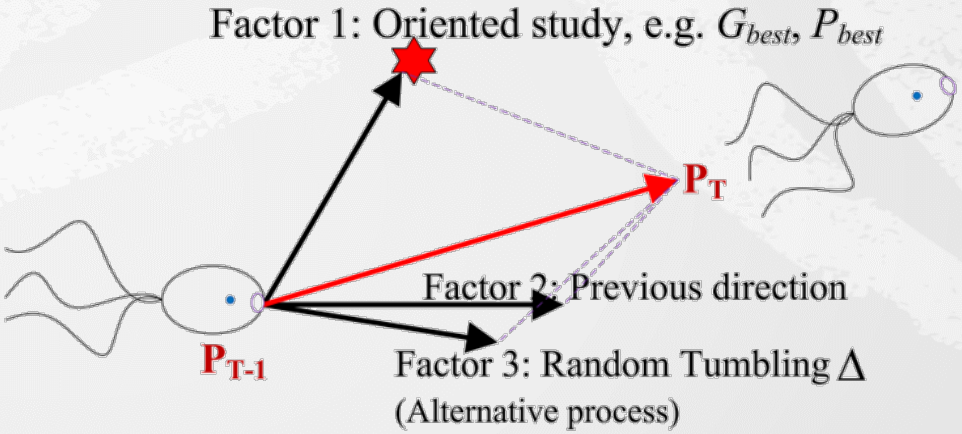
$$C(i) = C_{\min} + \left(\frac{\text{iter}_{\max} - \text{iter}_j}{\text{iter}_{\max}} \right)^n (C_{\max} - C_{\min})$$

Bacterial Colony Optimization

Information transfer mechanism:

$$\text{Poistion}_i(T) = \text{Poistion}_i(T-1) + C(i) * \Big[f_i * (G_{\text{best}} - \text{Position}_i(T-1)) + (1 - f_i) * (P_{\text{best}_i} - \text{Position}_i(T-1)) + \text{turbulent}_i \Big]$$





Bacterial Colony Optimization

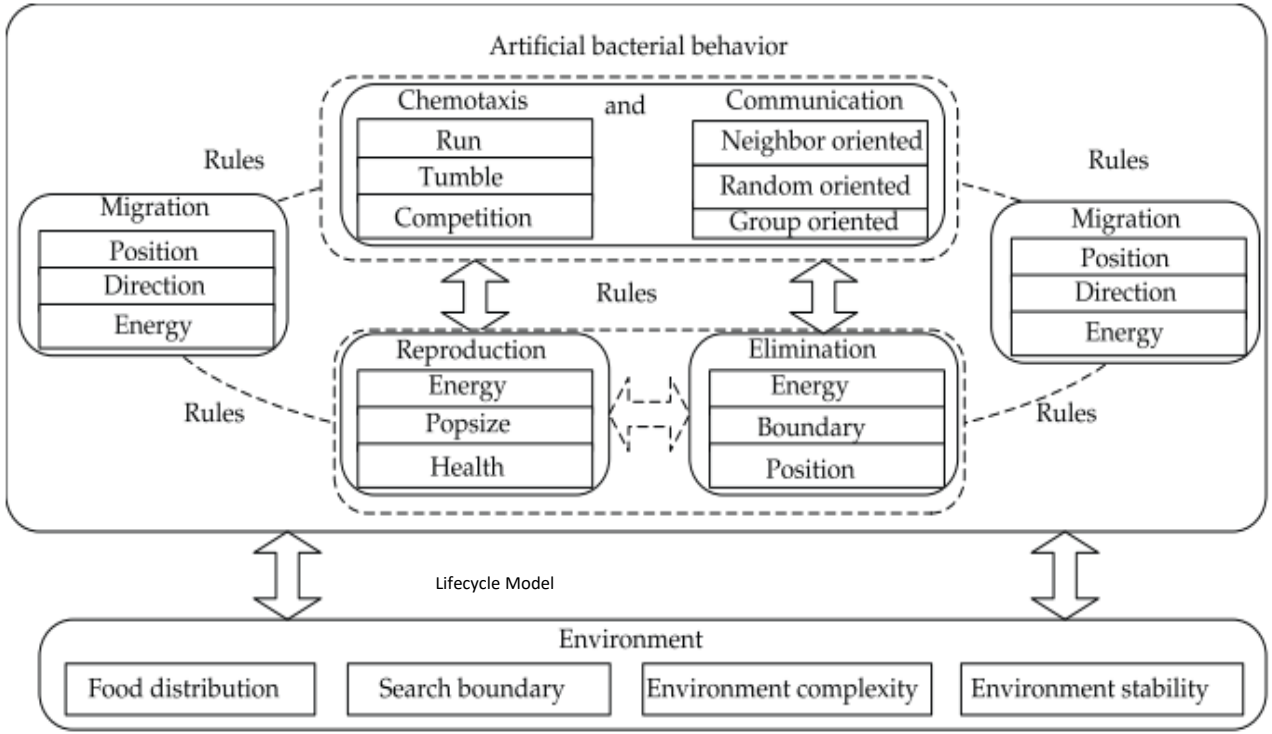


Figure 3: Lifecycle model.

Multi-objective optimization based on Dominance

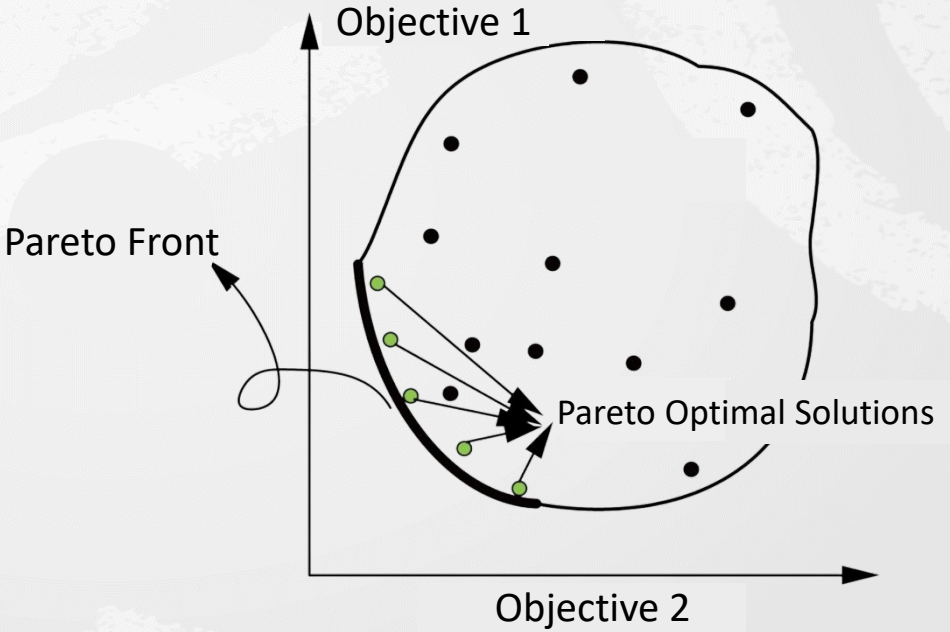
Equilibrium solution set: Pareto Front

A dominate B:

A: (90, 10%)
B: (70, 30%)

C, D do not dominate each other

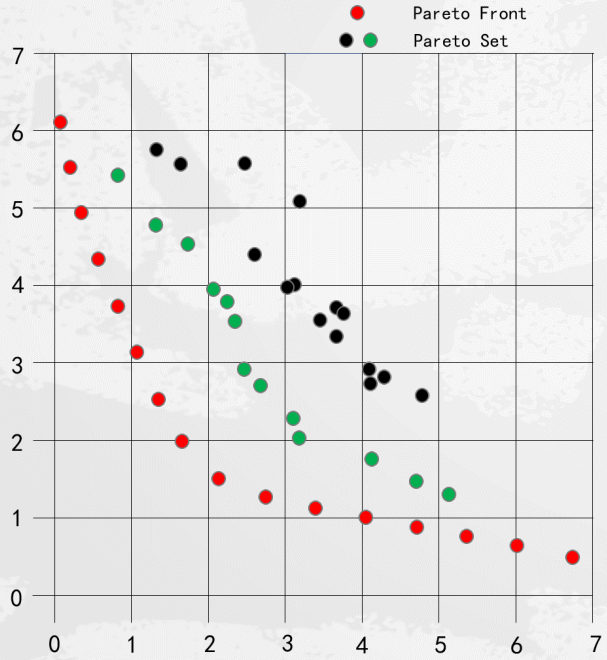
C: (50, 10%)
D: (80, 20%)



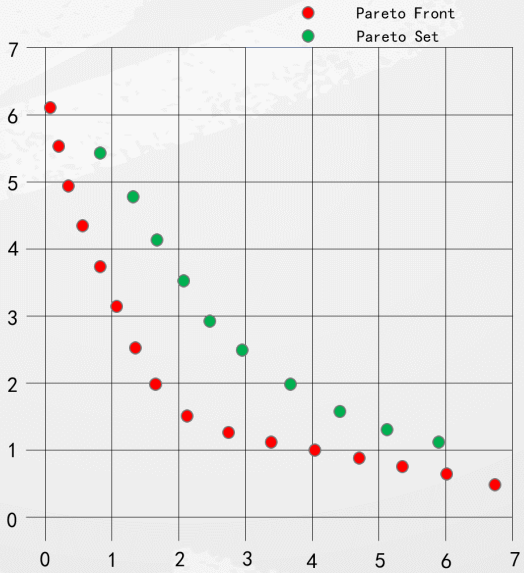
Multi-objective optimization based on Dominance

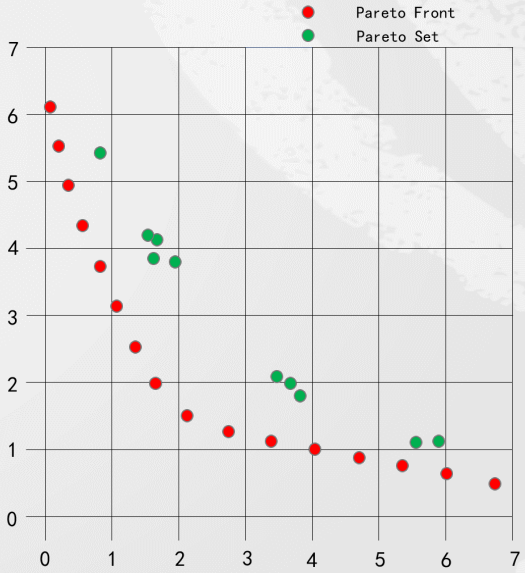
Pareto Optimal Solution Set

①: Convergence



②: Diversity (Spread)



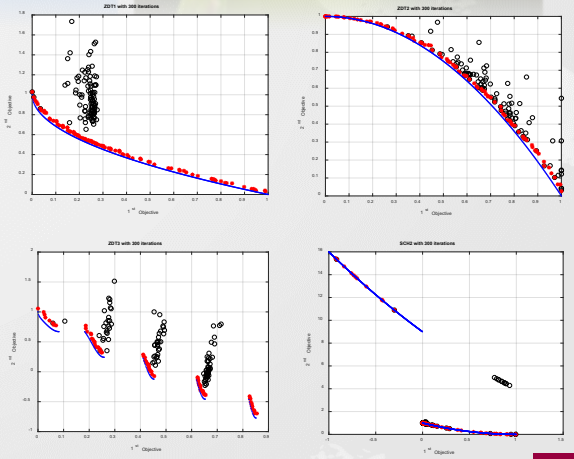


Multi-objective Bacterial Colony Optimization with Dynamic Multi-Leader Co-evolution

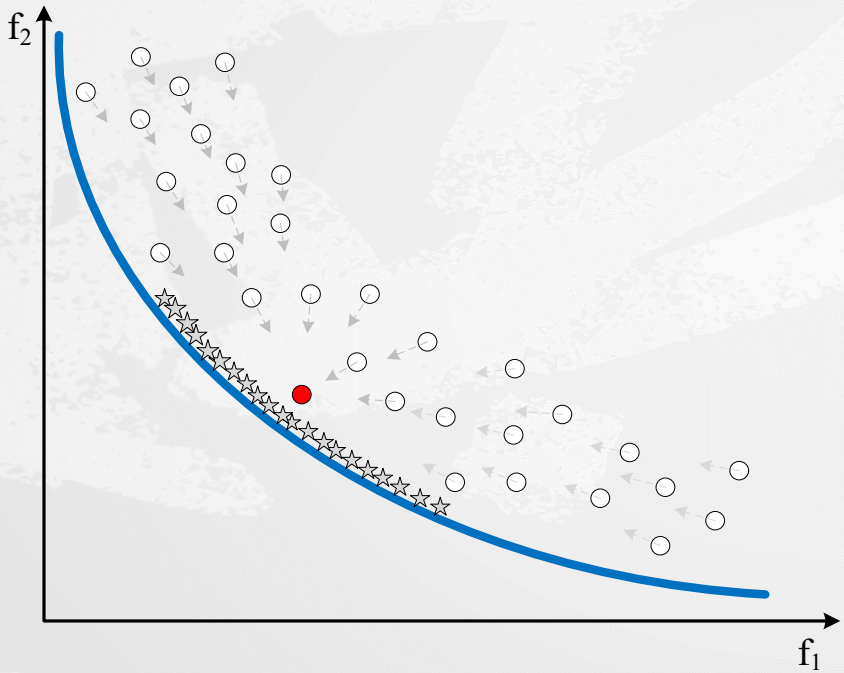
Multi-objective optimization based on Swarm Intelligence

1. **Computational Complexity**
2. **Insufficient convergence capacity**
3. **Lack of diversity**

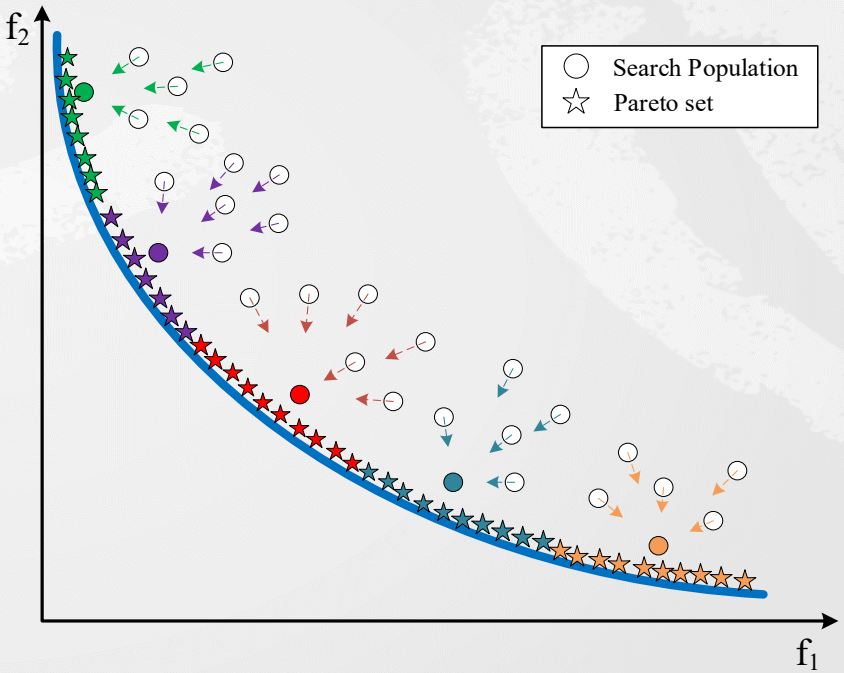
- Learning approach not adapted to multi-objective optimization problems
- Single approach to leadership selection and lack of robustness
- Inefficient and irrational mechanisms for information exchange between populations



Clustering-based Dynamic Multi-leaders Selection Mechanism



(a)



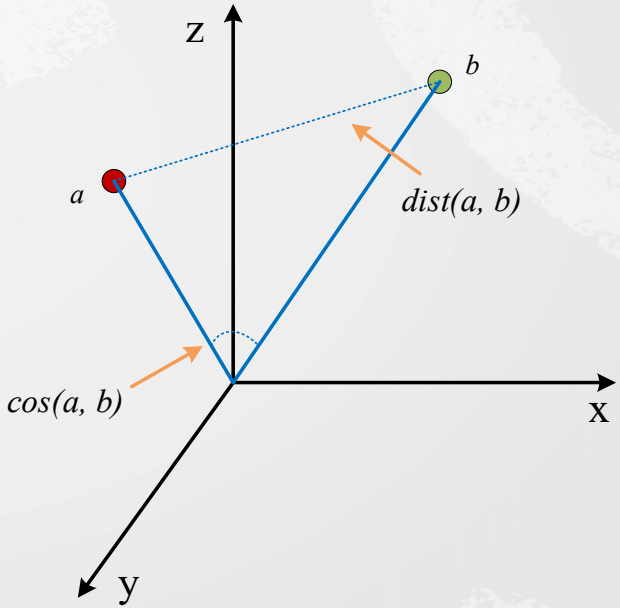
(b)

Clustering-based Dynamic Multi-leaders Selection Mechanism

- Evolutionary direction similarity assessment

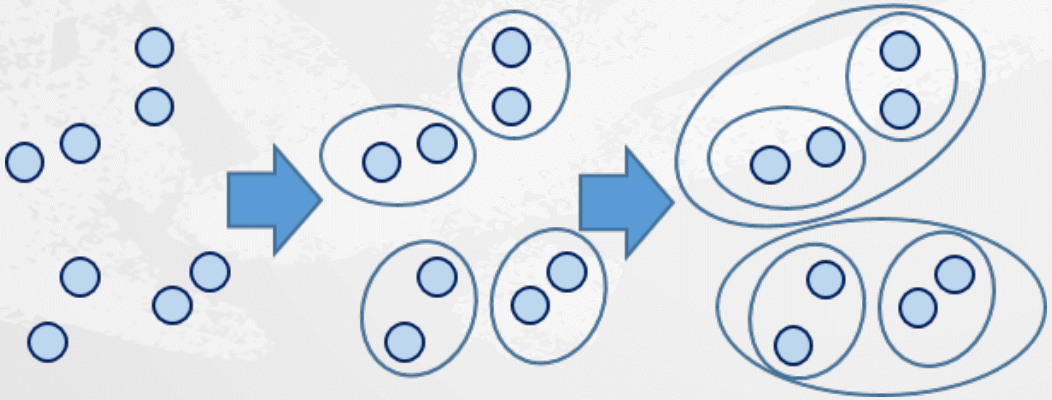
$$f_j^{'}(p) = \frac{f_j(p) - z_j^*}{z_j^- - z_j^*} \in [0,1]$$

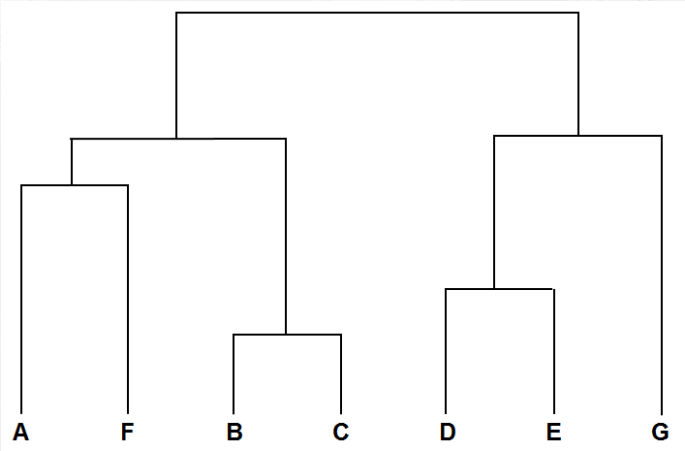
$$\cos Sim[a,b] = \frac{\sum_{i=1}^{m}(a_i \times b_i)}{\sqrt{\sum_{i=1}^{m}(a_i)^2} \times \sqrt{\sum_{i=1}^{m}(b_i)^2}}$$



Clustering-based Dynamic Multi-leaders Selection Mechanism

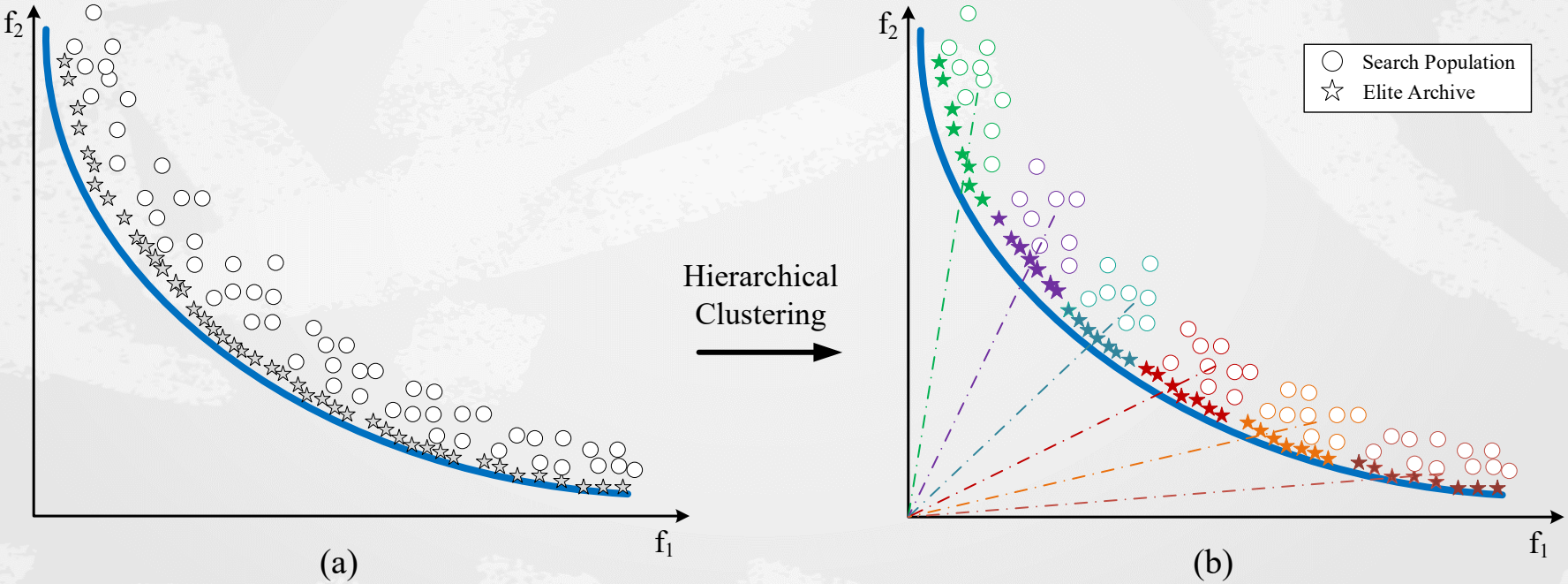
- Cohesive hierarchical clustering





Clustering-based Dynamic Multi-leaders Selection Mechanism

- Dynamic subpopulations



Clustering-based Dynamic Multi-leaders Selection Mechanism

- Leader selection

Ideal point distance: $SD[p]=\sqrt{\sum_{j}^{m}\left(p_j-O_j\right)^2}$

Crowding distance: $CrowdDis[p]=\sum_{j}^{m}\left(CrowdDis_j[p]+\frac{f_j(p)-z_j^*}{z_j^-z_j^*}\right)$

- The point with the closest distance to the ideal point is used as the optimal point for convergence:

Convergence leaders: $cleader[c]=min\left(SD[c]\right)$

- The point with the largest crowding distance is used as the optimal point for diversity:

Diversity leaders: $dleader[c]=max\left(CrowdDis[c]\right)$

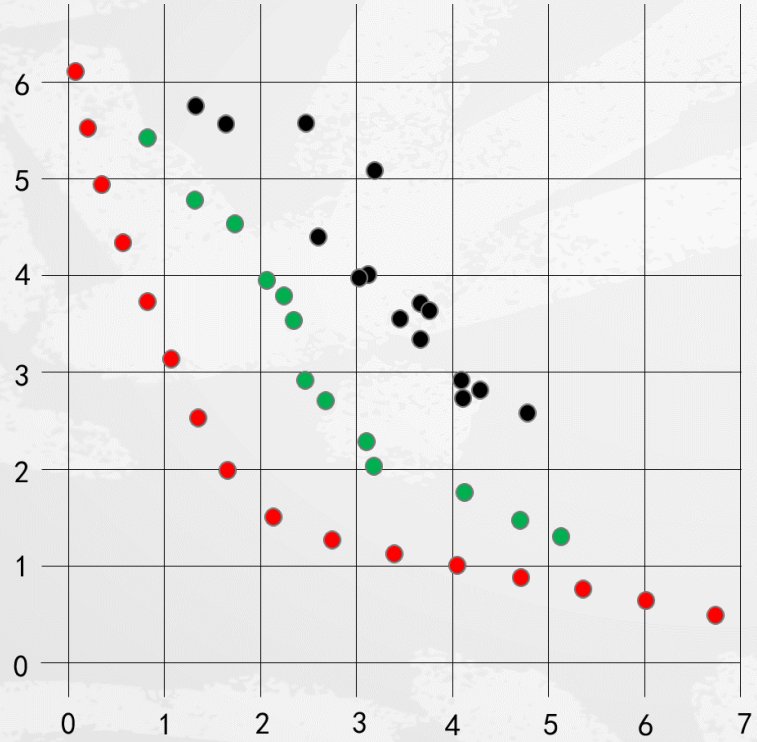
Clustering-based Dynamic Multi-leaders Selection Mechanism

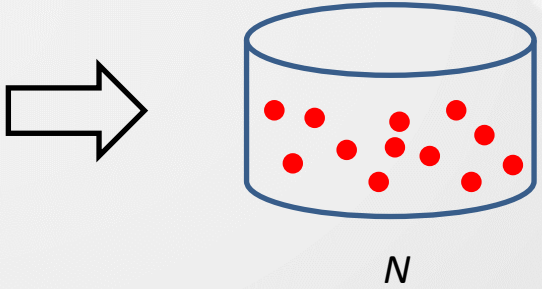
Algorithm 1. Dynamic multi-leader selection mechanism

```
Input: Population P, Elite archive EA, cluNum
Output: cleader, dleader
Calculate the best value of each objective among U (U = P ∪ EA), z*
Calculate the worst value of each objective among U (U = P ∪ EA), z⁻
Normalize the whole population into [0,1] by Eq.(4), nor_S
for i = 1 to |U| -1
    for j = 2 to |U|
        Calculate cosine distance between two individuals in nor_S by Eq.(5), cosSim[i, j]
    end
end
cluster = HClustering(nor_S, cosSim, cluNum)
    /* hierarchical clustering based on direction, return cluster label of every individual*/
for p ∈ P
    Calculate the Euclidean distances of solutions P to the origin O, SD[p]
end
for q ∈ EA
    Calculate the Euclidean distances of solutions in EA to the origin O by Eq.(6), SD[q]
    Calculate the crowding distance in EA by Eq.(7), CrowdDis[q]
end
for c ∈ clusters        /* c is composed of a certain number of searching individuals */
    find the convergence leader cleader in clusters by Eq.(8)
    find the diversity leader dleader in clusters by Eq.(9)
end
Return cleader, dleader
```

Elite archive evolution

- Elite archive





Elite archive evolution

Algorithm 2. Elite archive evolution

```
Input: Elite archive EA, N, offspring = [ ], Matching pool MP = [ ]
Output: Elite archive EA
if |EA| <= N/8
    Updated elite archive EA = EA
elseif N/8 <= |EA| <= N/5
    for q ∈ EA
        Calculate the crowding distance in EA by Eq.(7)
        Repeat each individual by Eq.(10)
        Add duplicated group into MP
    end
else |EA| >= N/5
    for q ∈ EA
        Calculate the crowding distance in EA by Eq.(7)
        Sort CrowdDis of each individual with ascending order
        Repeat the first N/5 individuals by Eq.(10)
        Add duplicated group into MP
    end
end
for i = 1 to N
    p1 = MP[i], randomly choose a spouse p2 from MP
    u = SBX(p1, p2)
    v = PM(v)
    offspring = offspring ∪ v
end
EA = updatingEA(EA,offspring)
Return EA
```

$$CrowdDis[p] = \sum_{j}^{m}\left(CrowdDis_j[p] + \frac{f_j(p) - z_j^*}{z_j^- - z_j^*} \right)$$

$$duplicatedNum_p = \left\lceil N \cdot \frac{CrowdDis[p]}{\sum_{p}^{n} CrowdDis[p]} \right\rceil$$

The Adaptive Bacterial Learning Optimization Behaviors

- Tumble:

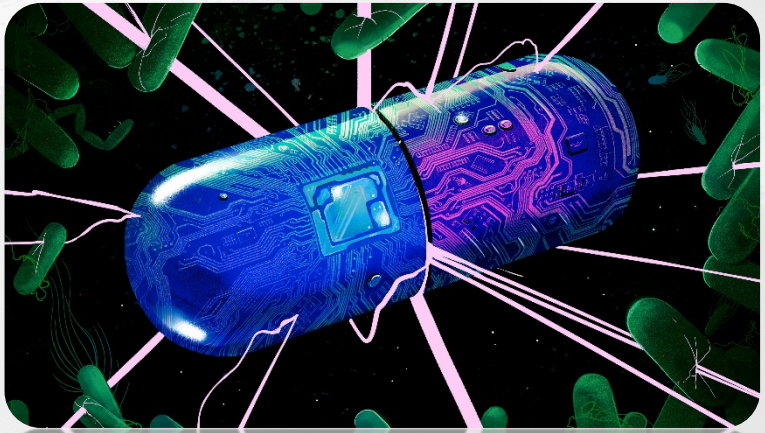
$$Tumble = C_i \cdot \{r_c \cdot (cleader - p(t)) + r_d \cdot (dleader - p(t))\}$$

- Learning:

$$p(t) = w \cdot p(t-1) + Tumble$$

- Adaptive elimination:

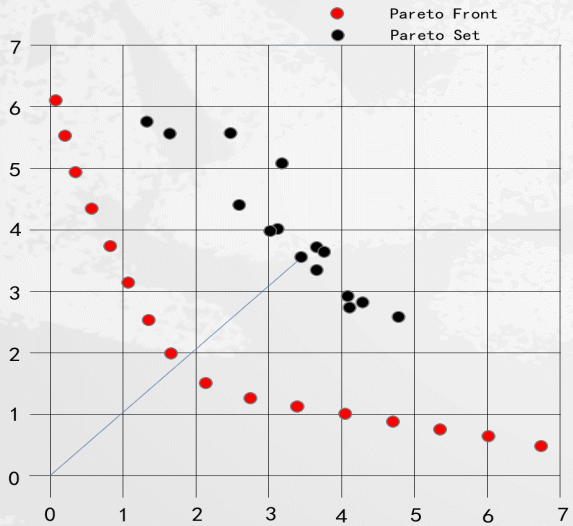
$$Ped = -2counter^{-1} + 1 \quad counter \geq 3$$



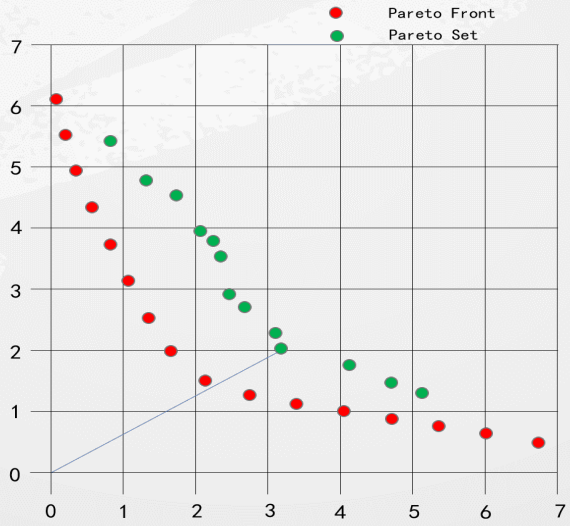
The Adaptive Bacterial Learning Optimization Behaviors

- Adaptive elimination:

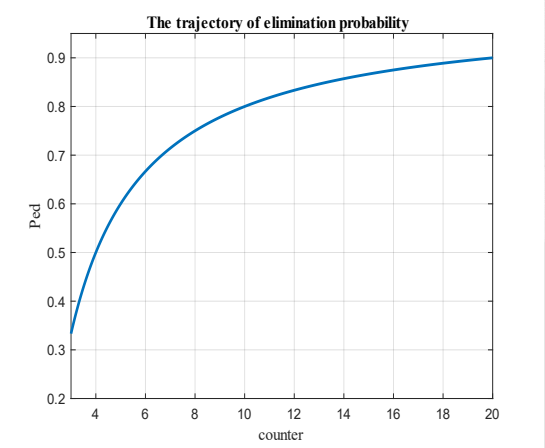
$$Ped = -2counter^{-1} + 1 \quad counter \geq 3$$



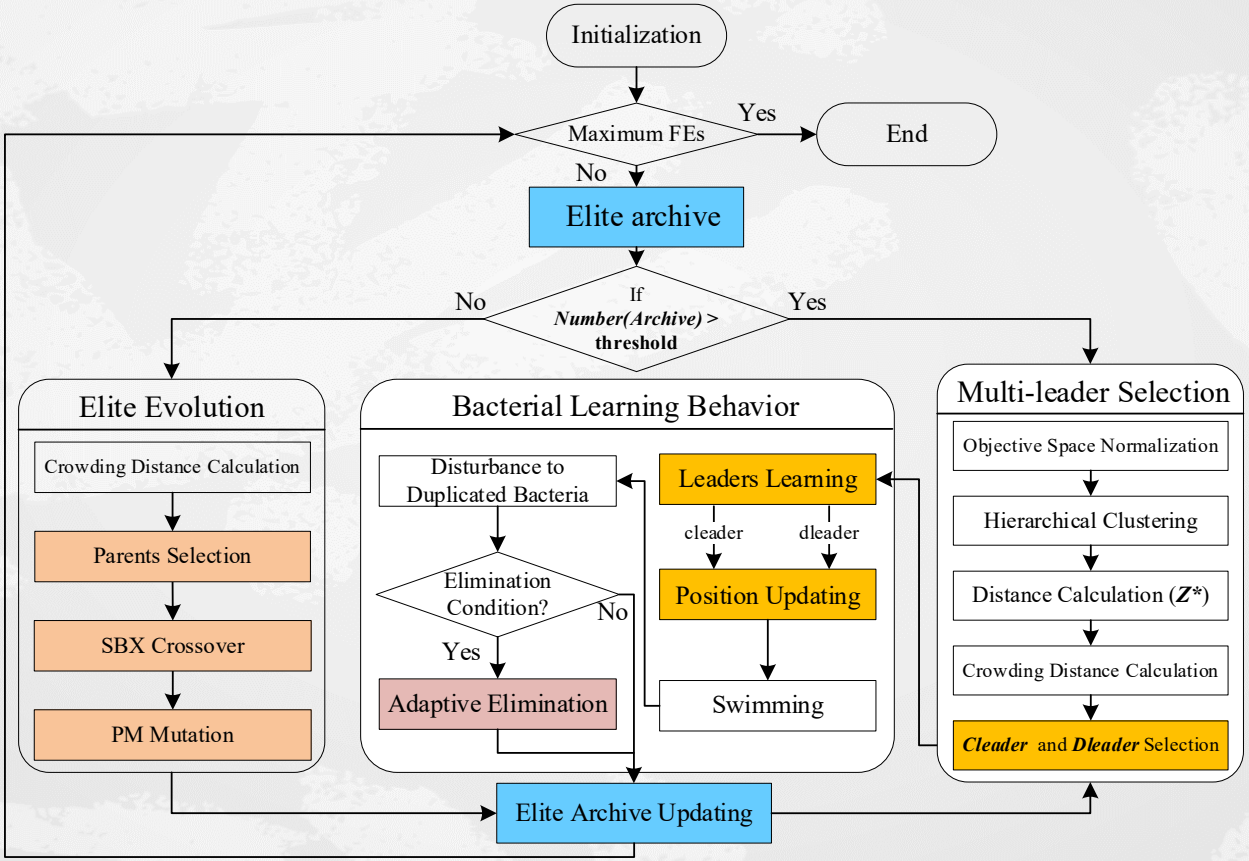
Last iteration



Current iteration



Multi-objective Bacterial Colony Optimization with Dynamic Multi-Leader Co-evolution



Experiments

Table 3. HV values of the proposed MBCO/DML and three state-of-the-art MOPSOs on 24 selected benchmarks

Problem	MOPSO	dMOPSO	NMPSO	MBCO/DML
FON	6.3316e-1 (3.92e-3)	**6.4165e-1 (3.34e-5)**	6.4017e-1 (1.38e-4)	6.4073e-1 (1.11e-4)
KUR	**7.1578e-1 (1.37e-2)**	7.1320e-1 (2.54e-4)	6.7701e-1 (9.17e-3)	7.1474e-1 (1.60e-4)
POL	1.0818e+0 (5.86e-4)	1.0573e+0 (8.95e-5)	1.0307e+0 (3.04e-2)	**1.0830e+0 (2.23e-5)**
SCH1	1.0668e+0 (5.56e-4)	1.0555e+0 (4.12e-6)	1.0329e+0 (1.28e-2)	**1.0693e+0 (4.16e-5)**
SCH2	8.8199e-1 (5.55e-3)	**1.1922e+0 (4.27e-5)**	8.7505e-1 (4.35e-3)	8.8105e-1 (2.46e-5)
ZDT1	9.0301e-2 (8.93e-2)	9.1569e-1 (3.55e-3)	6.9151e-1 (1.20e-2)	**9.3052e-1 (4.14e-5)**
ZDT2	2.6862e-4 (1.47e-3)	6.0599e-1 (1.08e-1)	4.3567e-1 (2.12e-3)	**6.5511e-1 (4.20e-5)**
ZDT3	1.3521e-1 (1.07e-1)	8.0511e-1 (9.14e-4)	5.6681e-1 (2.98e-4)	**8.0985e-1 (1.91e-5)**
ZDT4	0.0000e+0 (0.00e+0)	9.2623e-1 (1.48e-3)	6.8663e-1 (2.35e-2)	**9.3035e-1 (2.72e-4)**
ZDT6	3.5685e-1 (8.58e-2)	**5.7338e-1 (3.22e-6)**	3.8777e-1 (3.12e-4)	5.7290e-1 (3.17e-4)
F1	5.5289e-1 (2.26e-2)	9.2864e-1 (2.09e-5)	7.0192e-1 (7.10e-3)	**9.2905e-1 (1.95e-4)**
F2	1.9274e-1 (1.67e-2)	**6.5478e-1 (6.37e-5)**	2.0576e-1 (1.65e-1)	6.5306e-1 (2.82e-4)
F3	0.0000e+0 (0.00e+0)	2.3733e-1 (1.22e-2)	2.2961e-1 (1.15e-2)	**2.3854e-1 (1.64e-2)**
F5	5.0351e-1 (1.30e-2)	9.1944e-1 (2.08e-3)	6.9612e-1 (8.62e-3)	**9.2418e-1 (6.33e-4)**
F6	1.7696e-1 (1.25e-2)	5.0857e-1 (1.12e-1)	4.3458e-1 (2.45e-3)	**6.4599e-1 (1.14e-3)**
F7	0.0000e+0 (0.00e+0)	2.1000e-1 (6.64e-8)	1.6032e-1 (5.88e-2)	**2.1019e-1 (7.32e-4)**
F9	2.9631e-4 (1.19e-3)	8.1787e-1 (3.77e-2)	6.7193e-1 (2.45e-2)	**9.0843e-1 (1.45e-2)**
F10	0.0000e+0 (0.00e+0)	2.5366e-1 (3.47e-3)	1.8714e-1 (1.70e-2)	**3.2658e-1 (1.04e-2)**
F4	3.5064e-1 (1.85e-1)	8.3765e-1 (3.27e-2)	5.6220e-1 (1.13e-3)	**8.4616e-1 (6.07e-3)**
F8	1.0867e-1 (9.91e-2)	7.3404e-1 (1.46e-2)	4.1683e-1 (3.86e-2)	**7.6740e-1 (3.53e-2)**
DTLZ4	7.2792e-1 (7.14e-2)	7.7844e-1 (1.91e-2)	**8.8439e-1 (5.25e-2)**	8.5600e-1 (5.87e-3)
DTLZ5	3.8761e-1 (5.26e-3)	3.5369e-1 (9.80e-3)	3.9531e-1 (1.17e-3)	**4.0051e-1 (8.11e-5)**
DTLZ6	2.6515e-2 (1.01e-1)	3.7906e-1 (1.19e-5)	3.9592e-1 (1.06e-3)	**4.0075e-1 (3.41e-5)**
DTLZ7	4.7040e-2 (7.22e-2)	4.5785e-1 (7.48e-3)	**4.8886e-1 (1.99e-3)**	4.7680e-1 (4.61e-3)
Best/All	1/24	4/24	2/24	17/24

Experiments

Table 6. HV values of the proposed MBCO/DML and three novel MOBAs on 24 selected benchmarks

Problem	MOBFO	MORBCO	MCMBFO	MBCO/DML
FON	3.2092e-1 (7.00e-2)	6.3504e-1 (1.36e-3)	6.3077e-1 (3.08e-3)	**6.4073e-1 (1.11e-4)**
KUR	3.3698e-1 (2.37e-1)	7.0380e-1 (2.73e-3)	**7.4248e-1 (1.28e-2)**	7.1474e-1 (1.60e-4)
POL	1.0743e+0 (6.90e-3)	1.0786e+0 (3.74e-3)	1.0779e+0 (3.74e-3)	**1.0830e+0 (2.23e-5)**
SCH1	1.0157e+0 (2.32e-2)	1.0653e+0 (9.60e-4)	1.0692e+0 (2.43e-4)	**1.0693e+0 (4.16e-5)**
SCH2	8.5935e-1 (2.04e-2)	8.7901e-1 (2.23e-3)	**8.8151e-1 (4.15e-3)**	8.8105e-1 (2.46e-5)
ZDT1	5.3800e-1 (2.38e-2)	9.1830e-1 (4.19e-3)	9.1220e-1 (4.46e-3)	**9.3052e-1 (4.14e-5)**
ZDT2	2.1000e-1 (1.41e-16)	5.5639e-1 (1.72e-1)	6.0669e-1 (9.47e-2)	**6.5511e-1 (4.20e-5)**
ZDT3	2.1000e-1 (5.53e-10)	8.0814e-1 (3.10e-2)	8.0546e-1 (1.79e-3)	**8.0985e-1 (1.91e-5)**
ZDT4	4.6741e-1 (1.08e-2)	8.6431e-1 (1.37e-1)	0.0000e+0 (0.00e+0)	**9.3035e-1 (2.72e-4)**
ZDT6	2.3640e-1 (1.06e-3)	5.5343e-1 (1.34e-2)	5.4140e-1 (9.18e-2)	**5.7290e-1 (3.17e-4)**
F1	2.1000e-1 (7.04e-10)	9.2553e-1 (8.52e-4)	5.3494e-1 (3.21e-2)	**9.2905e-1 (1.95e-4)**
F2	**6.8794e-1 (8.85e-3)**	6.4985e-1 (9.52e-4)	2.0991e-1 (5.98e-5)	6.5306e-1 (2.82e-4)
F3	2.2991e-1 (8.58e-3)	**5.6509e-1 (3.43e-3)**	0.0000e+0 (0.00e+0)	2.3854e-1 (1.64e-2)
F5	6.0242e-1 (8.52e-3)	9.1479e-1 (5.38e-3)	4.7609e-1 (2.87e-3)	**9.2418e-1 (6.33e-4)**
F6	6.0852e-1 (5.16e-3)	6.4218e-1 (3.24e-3)	2.4233e-1 (1.18e-3)	**6.4599e-1 (1.14e-3)**
F7	**8.8902e-1 (2.60e-2)**	2.7776e-1 (5.83e-2)	0.0000e+0 (0.00e+0)	2.1019e-1 (7.32e-4)
F9	4.7601e-1 (9.71e-2)	**9.1830e-1 (1.78e-3)**	7.2100e-1 (1.02e-2)	9.0843e-1 (1.45e-2)
F10	**8.9310e-1 (6.08e-2)**	7.9401e-1 (1.55e-1)	2.2794e-1 (5.52e-2)	3.2658e-1 (1.04e-2)
F4	0.0000e+0 (0.00e+0)	**8.7282e-1 (5.76e-3)**	0.0000e+0 (0.00e+0)	8.4616e-1 (6.07e-3)
F8	5.6292e-1 (7.88e-3)	7.4309e-1 (3.34e-2)	5.9859e-1 (6.64e-2)	**7.6740e-1 (3.53e-2)**
DTLZ4	6.3052e-3 (2.74e-2)	3.1103e-1 (9.78e-2)	6.8930e-1 (2.74e-2)	**8.5600e-1 (5.87e-3)**
DTLZ5	1.1295e-2 (1.36e-2)	2.3256e-1 (8.47e-2)	8.2924e-2 (3.54e-3)	**4.0051e-1 (8.11e-5)**
DTLZ6	3.7440e-1 (3.20e-2)	1.9137e-1 (1.58e-1)	9.5088e-2 (2.03e-2)	**4.0075e-1 (3.41e-5)**
DTLZ7	3.9194e-1 (2.21e-2)	4.6882e-1 (7.07e-3)	4.4348e-1 (8.46e-3)	**4.7680e-1 (4.61e-3)**
Best/All	3/24	3/24	2/24	16/24

Experiments

Table 9. HV values of the proposed MBCO/DML and four competitive MOEAs on 24 selected benchmarks

Problem	eMOEA	BiGE	RVEA	MOCell	MBCO/DML
FON	6.1259e-1 (2.14e-3)	6.2317e-1 (2.25e-2)	6.3908e-1 (1.24e-3)	6.3861e-1 (2.32e-4)	**6.4073e-1 (1.11e-4)**
KUR	7.1353e-1 (3.68e-4)	7.1285e-1 (1.34e-2)	7.0724e-1 (3.31e-3)	7.1389e-1 (2.04e-4)	**7.1474e-1 (1.60e-4)**
POL	1.0824e+0 (1.36e-4)	1.0734e+0 (6.07e-3)	1.0810e+0 (1.36e-4)	1.0827e+0 (6.86e-5)	**1.0830e+0 (2.23e-5)**
SCH1	1.0635e+0 (5.47e-4)	1.0651e+0 (9.36e-4)	1.0681e+0 (1.12e-5)	1.0687e+0 (1.50e-4)	**1.0693e+0 (4.16e-5)**
SCH2	8.7575e-1 (1.05e-3)	8.7878e-1 (1.77e-3)	8.7804e-1 (8.86e-5)	8.8073e-1 (8.65e-5)	**8.8105e-1 (2.46e-5)**
ZDT1	8.9603e-1 (2.79e-3)	9.1950e-1 (5.08e-3)	9.2676e-1 (9.94e-4)	9.2858e-1 (2.46e-4)	**9.3052e-1 (4.14e-5)**
ZDT2	6.1963e-1 (3.55e-3)	6.4563e-1 (3.52e-3)	6.5191e-1 (1.77e-3)	6.5357e-1 (2.07e-4)	**6.5511e-1 (4.20e-5)**
ZDT3	7.8759e-1 (2.63e-3)	8.0585e-1 (7.77e-4)	8.0485e-1 (1.04e-3)	**8.1214e-1 (1.62e-2)**	8.0985e-1 (1.91e-5)
ZDT4	8.1584e-1 (5.51e-2)	9.1162e-1 (7.02e-3)	8.5637e-1 (3.62e-2)	9.2147e-1 (2.64e-3)	**9.3035e-1 (2.72e-4)**
ZDT6	5.2502e-1 (4.94e-3)	5.6776e-1 (9.72e-4)	5.1087e-1 (1.02e-2)	5.6871e-1 (6.75e-4)	**5.7290e-1 (3.17e-4)**
F1	7.1011e-1 (2.65e-2)	7.2388e-1 (2.52e-2)	7.7381e-1 (2.17e-2)	7.9596e-1 (3.16e-2)	**9.2905e-1 (1.95e-4)**
F2	2.5424e-1 (3.48e-2)	2.1633e-1 (1.11e-2)	3.0707e-1 (6.79e-2)	2.8864e-1 (7.71e-2)	**6.5306e-1 (2.82e-4)**
F3	2.0119e-3 (1.10e-2)	1.7606e-2 (3.58e-2)	0.0000e+0 (0.00e+0)	1.1951e-2 (3.10e-2)	**2.3854e-1 (1.64e-2)**
F5	6.3211e-1 (2.91e-2)	6.3230e-1 (8.44e-3)	6.6574e-1 (2.85e-2)	6.7576e-1 (1.92e-2)	**9.2418e-1 (6.33e-4)**
F6	3.0431e-1 (1.57e-2)	3.7811e-1 (6.71e-3)	3.1544e-1 (9.45e-3)	3.2861e-1 (1.63e-2)	**6.4599e-1 (1.14e-3)**
F7	0.0000e+0 (0.00e+0)	8.7186e-2 (2.44e-2)	0.0000e+0 (0.00e+0)	8.1764e-2 (3.29e-2)	**2.1019e-1 (7.32e-4)**
F9	8.2203e-1 (2.10e-2)	8.5685e-1 (2.53e-2)	8.5043e-1 (1.65e-2)	8.9657e-1 (1.14e-2)	**9.0843e-1 (1.45e-2)**
F10	2.1164e-1 (4.96e-3)	2.4668e-1 (1.79e-3)	2.2073e-1 (7.49e-2)	2.4684e-1 (2.19e-3)	**3.2658e-1 (1.04e-2)**
F4	5.7494e-1 (4.59e-2)	7.2203e-1 (5.10e-2)	7.7934e-1 (1.77e-2)	7.2953e-1 (1.13e-2)	**8.4616e-1 (6.07e-3)**
F8	5.5270e-1 (3.18e-2)	6.6858e-1 (3.01e-2)	**7.8489e-1 (2.04e-2)**	7.1729e-1 (2.29e-2)	7.6740e-1 (3.53e-2)
DTLZ4	8.0810e-1 (1.83e-1)	8.5177e-1 (1.17e-1)	**8.8445e-1 (5.24e-2)**	7.7634e-1 (2.18e-1)	8.5600e-1 (5.87e-3)
DTLZ5	3.5996e-1 (2.30e-3)	3.9151e-1 (2.17e-3)	3.5400e-1 (1.21e-3)	3.9851e-1 (2.82e-4)	**4.0051e-1 (8.11e-5)**
DTLZ6	3.7154e-1 (1.12e-3)	2.3293e-1 (3.87e-3)	3.4221e-1 (1.25e-2)	4.0020e-1 (1.22e-4)	**4.0075e-1 (3.41e-5)**
DTLZ7	4.5300e-1 (4.34e-2)	4.4574e-1 (7.71e-2)	4.7258e-1 (3.54e-3)	4.6903e-1 (1.65e-2)	**4.7680e-1 (4.61e-3)**
Best/All	0/24	0/24	2/24	1/24	21/24

Experiments

Table 4. Spread values of the proposed MBCO/DML and three state-of-the-art MOPSOs on 24 selected benchmarks

Problem	MOPSO	dMOPSO	NMPSO	MBCO/DML
FON	7.7833e-1 (5.28e-2)	**1.5185e-1 (3.91e-4)**	3.4052e-1 (3.55e-2)	1.5492e-1 (1.26e-2)
KUR	6.9453e-1 (5.95e-2)	5.2582e-1 (4.47e-3)	1.8784e+0 (2.12e-2)	**2.0532e-1 (2.11e-2)**
POL	8.1469e-1 (5.76e-2)	9.2299e-1 (1.80e-3)	1.1960e+0 (8.94e-2)	**1.8655e-1 (2.01e-2)**
SCH1	7.8093e-1 (5.41e-2)	5.7827e-1 (1.92e-4)	1.4424e+0 (9.71e-2)	**1.7537e-1 (1.53e-2)**
SCH2	8.8586e-1 (6.24e-2)	9.5195e-1 (7.53e-3)	1.4303e+0 (1.51e-1)	**4.5059e-1 (2.39e-2)**
ZDT1	8.7842e-1 (3.69e-2)	2.8557e-1 (9.75e-3)	1.4471e+0 (8.67e-2)	**1.7281e-1 (1.62e-2)**
ZDT2	9.9209e-1 (2.36e-2)	2.0073e-1 (1.55e-1)	1.1816e+0 (6.60e-2)	**1.6079e-1 (1.80e-2)**
ZDT3	9.1583e-1 (3.55e-2)	5.4847e-1 (2.97e-2)	1.7655e+0 (6.10e-3)	**2.1009e-1 (2.24e-2)**
ZDT4	9.9148e-1 (8.76e-3)	2.7355e-1 (5.96e-3)	1.4135e+0 (1.07e-1)	**1.3962e-1 (1.49e-2)**
ZDT6	9.6748e-1 (1.91e-1)	**1.3654e-1 (6.76e-4)**	4.8094e-1 (6.20e-2)	7.9691e-1 (4.69e-1)
F1	8.7670e-1 (4.11e-2)	2.7364e-1 (8.39e-4)	1.2625e+0 (1.22e-1)	**1.6673e-1 (1.25e-2)**
F2	9.2750e-1 (2.54e-2)	**1.4315e-1 (1.31e-3)**	1.0474e+0 (8.18e-2)	1.6901e-1 (1.09e-2)
F3	9.7243e-1 (9.20e-3)	**5.5494e-1 (3.61e-2)**	1.1410e+0 (1.45e-1)	8.7042e-1 (1.06e-1)
F5	9.2043e-1 (3.25e-2)	2.5861e-1 (5.30e-3)	1.3737e+0 (1.10e-1)	**1.6015e-1 (1.32e-2)**
F6	9.4344e-1 (1.60e-2)	4.5666e-1 (2.54e-1)	1.1677e+0 (7.55e-2)	**1.7321e-1 (1.51e-2)**
F7	9.8675e-1 (5.73e-3)	**6.1055e-1 (5.04e-2)**	1.0127e+0 (1.74e-1)	9.5294e-1 (1.06e-1)
F9	9.3468e-1 (2.53e-2)	5.2699e-1 (7.39e-2)	1.5472e+0 (8.89e-2)	**1.8046e-1 (2.47e-2)**
F10	9.9848e-1 (3.26e-3)	9.9800e-1 (4.07e-4)	1.0171e+0 (9.11e-3)	**9.5858e-1 (5.52e-3)**
F4	4.2686e-1 (4.69e-2)	**2.0012e-1 (2.99e-2)**	3.3213e-1 (2.15e-2)	5.8167e-1 (2.24e-1)
F8	4.7748e-1 (5.17e-2)	**3.8345e-1 (9.06e-2)**	1.1369e+0 (2.21e-1)	4.9563e-1 (2.51e-1)
DTLZ4	6.4419e-1 (1.50e-1)	6.6424e-1 (1.27e-1)	**3.4291e-1 (1.23e-1)**	5.5522e-1 (5.99e-2)
DTLZ5	8.6869e-1 (5.59e-2)	1.6141e+0 (6.60e-2)	1.0852e+0 (9.66e-2)	**2.4243e-1 (2.11e-2)**
DTLZ6	6.0199e-1 (1.37e-1)	1.9794e+0 (2.13e-2)	1.0941e+0 (1.15e-1)	**2.3355e-1 (1.63e-2)**
DTLZ7	8.0369e-1 (1.51e-1)	1.0850e+0 (2.85e-2)	5.6031e-1 (3.75e-2)	**5.1079e-1 (4.28e-2)**
Best/All	0/24	7/24	1/24	16/24

Experiments

Table 7. Spread values of the proposed MBCO/DML and three novel MOBAs on 24 selected benchmarks

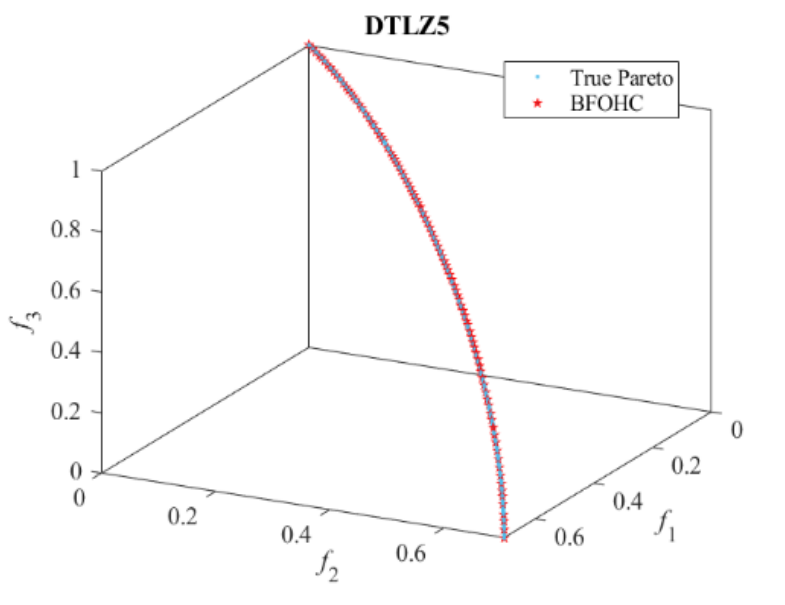
Problem	MOBFO	MORBCO	MCMBFO	MBCO/DML
FON	9.3634e-1 (2.45e-1)	1.0538e+0 (9.37e-2)	2.2344e-1 (1.93e-2)	**1.5492e-1 (1.26e-2)**
KUR	1.0136e+0 (2.27e-1)	1.3594e+0 (6.84e-2)	4.7291e-1 (7.09e-2)	**2.0532e-1 (2.11e-2)**
POL	1.0956e+0 (2.80e-1)	1.3110e+0 (8.45e-2)	7.2727e-1 (2.77e-1)	**1.8655e-1 (2.01e-2)**
SCH1	9.3160e-1 (2.00e-1)	1.0692e+0 (8.57e-2)	2.0708e-1 (2.64e-2)	**1.7537e-1 (1.53e-2)**
SCH2	1.0131e+0 (1.89e-1)	1.1740e+0 (8.98e-2)	3.8276e-1 (1.80e-1)	**4.5059e-1 (2.39e-2)**
ZDT1	1.1517e+0 (6.58e-2)	1.3301e+0 (1.81e-1)	2.2303e-1 (3.09e-2)	**1.7281e-1 (1.62e-2)**
ZDT2	1.0000e+0 (0.00e+0)	1.4118e+0 (3.38e-1)	3.2844e-1 (1.67e-1)	**1.6079e-1 (1.80e-2)**
ZDT3	1.1358e+0 (1.76e-1)	1.5807e+0 (2.09e-1)	2.8625e-1 (3.53e-2)	**2.1009e-1 (2.24e-2)**
ZDT4	1.0023e+0 (4.47e-2)	1.3033e+0 (1.25e-1)	1.0019e+0 (2.40e-2)	**1.3962e-1 (1.49e-2)**
ZDT6	1.0282e+0 (1.42e-2)	1.7759e+0 (1.49e-1)	**7.5600e-1 (4.37e-1)**	7.9691e-1 (4.69e-1)
F1	9.0514e-1 (1.14e-1)	9.7414e-1 (7.71e-2)	1.0132e+0 (1.24e-1)	**1.6673e-1 (1.25e-2)**
F2	9.0370e-1 (7.32e-2)	9.8729e-1 (9.08e-2)	1.0000e+0 (8.52e-4)	**1.6901e-1 (1.09e-2)**
F3	1.0114e+0 (6.53e-3)	1.1242e+0 (1.22e-1)	9.3745e-1 (2.85e-2)	**8.7042e-1 (1.06e-1)**
F5	1.6272e+0 (1.10e-1)	1.2784e+0 (1.13e-1)	8.6152e-1 (9.78e-3)	**1.6015e-1 (1.32e-2)**
F6	1.6337e+0 (1.68e-1)	1.2495e+0 (7.23e-2)	9.1809e-1 (9.16e-3)	**1.7321e-1 (1.51e-2)**
F7	**6.6209e-1 (2.64e-1)**	1.3728e+0 (1.13e-1)	8.9084e-1 (2.61e-2)	9.5294e-1 (1.06e-1)
F9	1.0711e+0 (1.51e-1)	1.2789e+0 (9.08e-2)	7.7150e-1 (1.02e-1)	**1.8046e-1 (2.47e-2)**
F10	1.2189e+0 (1.29e-1)	1.3178e+0 (2.02e-1)	9.9595e-1 (1.63e-3)	**9.5858e-1 (5.52e-3)**
F4	1.0097e+0 (4.89e-2)	8.0502e-1 (1.73e-1)	6.9994e-1 (1.77e-1)	**5.8167e-1 (2.24e-1)**
F8	1.3870e+0 (4.67e-1)	1.5254e+0 (2.37e-1)	1.1845e+0 (2.46e-1)	**4.9563e-1 (2.51e-1)**
DTLZ4	1.3069e+0 (1.63e-1)	1.2362e+0 (9.55e-2)	**4.8899e-1 (8.32e-2)**	5.5522e-1 (5.99e-2)
DTLZ5	6.4469e-1 (5.63e-2)	1.3083e+0 (1.36e-1)	6.3678e-1 (5.07e-2)	**2.4243e-1 (2.11e-2)**
DTLZ6	1.0409e+0 (5.51e-1)	1.4784e+0 (4.56e-1)	8.4543e-1 (2.11e-1)	**2.3355e-1 (1.63e-2)**
DTLZ7	1.1642e+0 (1.04e-1)	1.2902e+0 (1.17e-1)	**4.5037e-1 (2.90e-2)**	5.1079e-1 (4.28e-2)
Best/All	1/24	0/24	3/24	20/24

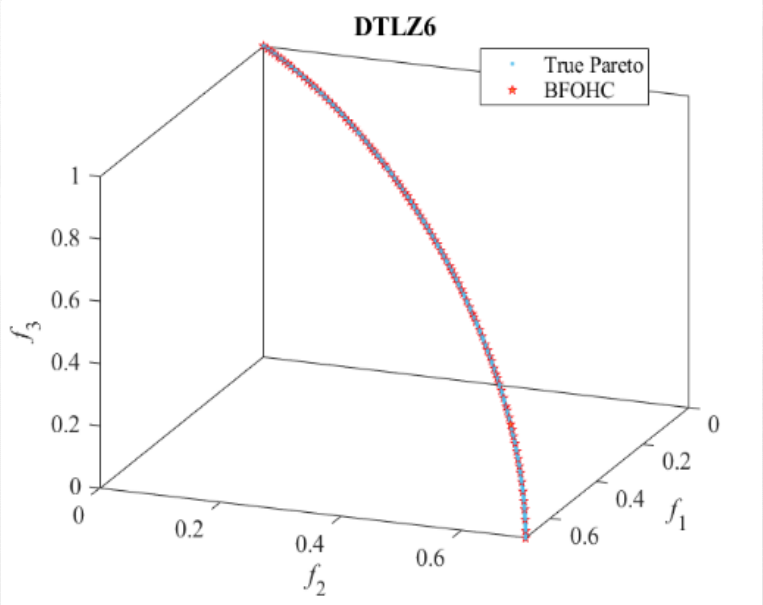
Experiments

Table 10. Spread values of the proposed MBCO/DML and four competitive MOEAs on 24 selected benchmarks

Problem	eMOEA	BiGE	RVEA	MOCell	MBCO/DML
FON	3.4758e-1 (4.06e-2)	1.2565e+0 (2.15e-1)	1.5754e-1 (1.17e-2)	3.9661e-1 (4.97e-2)	**1.5492e-1 (1.26e-2)**
KUR	3.9622e-1 (4.96e-2)	1.2636e+0 (1.72e-1)	3.3203e-1 (8.92e-2)	4.3646e-1 (4.74e-2)	**2.0532e-1 (2.11e-2)**
POL	7.3338e-1 (6.24e-2)	1.1439e+0 (1.09e-1)	1.0120e+0 (1.62e-2)	4.2048e-1 (5.80e-2)	**1.8655e-1 (2.01e-2)**
SCH1	5.5762e-1 (7.63e-2)	8.4813e-1 (7.69e-2)	7.0122e-1 (1.30e-3)	3.7669e-1 (4.85e-2)	**1.7537e-1 (1.53e-2)**
SCH2	8.0102e-1 (6.57e-2)	1.0720e+0 (5.37e-2)	**1.1288e-1 (5.52e-4)**	5.6184e-1 (4.92e-2)	4.5059e-1 (2.39e-2)
ZDT1	2.9531e-1 (7.55e-2)	1.0553e+0 (7.32e-2)	2.9734e-1 (1.07e-2)	3.2887e-1 (4.11e-2)	**1.7281e-1 (1.62e-2)**
ZDT2	3.0230e-1 (1.00e-1)	1.0865e+0 (7.81e-2)	**1.5621e-1 (2.51e-2)**	3.7034e-1 (4.54e-2)	1.6079e-1 (1.80e-2)
ZDT3	7.7202e-1 (1.06e-1)	1.1402e+0 (7.41e-2)	2.7290e-1 (2.13e-2)	3.8444e-1 (5.22e-2)	**2.1009e-1 (2.24e-2)**
ZDT4	1.4092e+0 (6.53e-1)	1.0445e+0 (1.24e-1)	4.9388e-1 (1.36e-1)	4.1596e-1 (4.96e-2)	**1.3962e-1 (1.49e-2)**
ZDT6	7.9726e-1 (4.29e-1)	1.0266e+0 (1.57e-1)	**2.6760e-1 (4.98e-2)**	3.7926e-1 (6.00e-2)	7.9691e-1 (4.69e-1)
F1	7.9728e-1 (1.34e-1)	1.0532e+0 (4.78e-2)	7.4996e-1 (4.76e-2)	6.8258e-1 (4.50e-2)	**1.6673e-1 (1.25e-2)**
F2	9.8369e-1 (8.76e-2)	1.0051e+0 (8.84e-3)	9.5625e-1 (6.74e-2)	9.5231e-1 (6.07e-2)	**1.6901e-1 (1.09e-2)**
F3	1.0133e+0 (7.22e-2)	1.0000e+0 (0.00e+0)	9.7612e-1 (2.12e-2)	1.0003e+0 (4.93e-4)	**8.7042e-1 (1.06e-1)**
F5	8.4534e-1 (6.33e-2)	1.0113e+0 (1.26e-2)	8.3716e-1 (3.36e-2)	7.8425e-1 (1.64e-2)	**1.6015e-1 (1.32e-2)**
F6	8.9922e-1 (5.39e-2)	1.0163e+0 (1.05e-2)	9.4188e-1 (3.14e-2)	8.7699e-1 (1.57e-2)	**1.7321e-1 (1.51e-2)**
F7	1.0373e+0 (1.31e-1)	1.0388e+0 (1.03e-1)	9.6485e-1 (1.91e-2)	1.0000e+0 (0.00e+0)	**9.5294e-1 (1.06e-1)**
F9	5.7219e-1 (1.07e-1)	1.0288e+0 (5.71e-2)	5.7133e-1 (6.21e-2)	4.6858e-1 (3.92e-2)	**1.8046e-1 (2.47e-2)**
F10	1.2206e+0 (7.32e-1)	1.0315e+0 (1.73e-1)	9.8402e-1 (4.17e-2)	9.9193e-1 (1.13e-3)	**9.5858e-1 (5.52e-3)**
F4	1.2748e+0 (3.54e-1)	1.5222e+0 (3.04e-1)	**2.2515e-1 (2.57e-2)**	8.1962e-1 (3.87e-1)	5.8167e-1 (2.24e-1)
F8	1.2658e+0 (2.49e-1)	1.8895e+0 (8.27e-2)	**2.2842e-1 (2.71e-2)**	8.6206e-1 (4.31e-1)	4.9563e-1 (2.51e-1)
DTLZ4	3.3936e-1 (1.74e-1)	8.9203e-1 (6.88e-2)	**1.9362e-1 (1.15e-1)**	5.2168e-1 (1.93e-1)	5.5522e-1 (5.99e-2)
DTLZ5	6.4163e-1 (6.48e-2)	1.1070e+0 (7.51e-2)	5.6981e-1 (7.83e-2)	3.9684e-1 (3.91e-2)	**2.4243e-1 (2.11e-2)**
DTLZ6	8.8859e-1 (3.88e-1)	1.0157e+0 (1.88e-2)	4.7718e-1 (8.27e-2)	5.4538e-1 (5.94e-2)	**2.3355e-1 (1.63e-2)**
DTLZ7	4.7772e-1 (1.05e-1)	9.8863e-1 (1.49e-1)	**3.6312e-1 (2.38e-2)**	4.7362e-1 (4.59e-2)	5.1079e-1 (4.28e-2)
Best/All	0/24	0/24	7/24	0/24	17/24

Experiments





THANKS

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